language queries (e.g. few-shot prompting (Brown et al., 2020)), encouraging reasoning traces (e.g. chain-of-thought (Wei et al., 2022)), or asking the model to iteratively refine its outputs (e.g., self-reflection (Shinn et al., 2024)). These leverage the language model's ability to learn in-context (Olsson et al., 2022) and can greatly improve its performance, robustness and reliability on many tasks.

Aider: An LLM-Based Coding Assistant. Our automated scientist directly implements ideas in code and uses a state-of-the-art open-source coding assistant, Aider (Gauthier, 2024). Aider is an agent framework that is designed to implement requested features, fix bugs, or refactor code in existing codebases. While Aider can in principle use any underlying LLM, with frontier models it achieves a remarkable success rate of 18.9% on the SWE Bench (Jimenez et al., 2024) benchmark, a collection of real-world GitHub issues. In conjunction with new innovations added in this work, this level of reliability enables us, for the first time, to fully automate the ML research process.

3. The AI Scientist

Overview. The AI Scientist has three main phases (Figure 1): (1) Idea Generation, (2) Experimental Iteration, and (3) Paper Write-up. After the write-up, we introduce and validate an LLM-generated review to assess the quality of the generated paper (Section 4). We provide The AI Scientist with a starting *code template* that reproduces a lightweight baseline training run from a popular model or benchmark. For example, this could be code that trains a small transformer on the works of Shakespeare (Karpathy, 2022), a classic proof-of-concept training run from natural language processing that completes within a few minutes. The AI Scientist is then free to explore any possible research direction. The template also includes a LaTeX folder that contains style files and section headers, along with simple plotting code. We provide further details on the templates in Section 6, but in general, each run starts with a representative small-scale experiment relevant to the topic area. The focus on small-scale experiments is not a fundamental limitation of our method, but simply for computational efficiency reasons and compute constraints on our end. We provide the prompts for all stages in Appendix A.

1. Idea Generation. Given a starting template, The AI Scientist first "brainstorms" a diverse set of novel research directions. We take inspiration from evolutionary computation and open-endedness research (Brant and Stanley, 2017; Lehman et al., 2008; Stanley, 2019; Stanley et al., 2017) and iteratively grow an archive of ideas using LLMs as the mutation operator (Faldor et al., 2024; Lehman et al., 2022; Lu et al., 2024b; Zhang et al., 2024). Each idea comprises a description, experiment execution plan, and (self-assessed) numerical scores of interestingness, novelty, and feasibility. At each iteration, we prompt the language model to generate an interesting new research direction conditional on the existing archive, which can include the numerical review scores from completed previous ideas. We use multiple rounds of chain-of-thought (Wei et al., 2022) and self-reflection (Shinn et al., 2024) to refine and develop each idea. After idea generation, we filter ideas by connecting the language model with the Semantic Scholar API (Fricke, 2018) and web access as a tool (Schick et al., 2024). This allows The AI Scientist to discard any idea that is too similar to existing literature.

2. Experiment Iteration. Given an idea and a template, the second phase of The AI Scientist first executes the proposed experiments and then visualizes its results for the downstream write-up. The AI Scientist uses Aider to first plan a list of experiments to run and then executes them in order. We make this process more robust by returning any errors upon a failure or time-out (e.g. experiments taking too long to run) to Aider to fix the code and re-attempt up to four times.

After the completion of each experiment, Aider is then given the results and told to take notes in the style of an experimental journal. Currently, it only conditions on text but in future versions, this could include data visualizations or any modality. Conditional on the results, it then re-plans and implements the next experiment. This process is repeated up to five times. Upon completion of

experiments, Aider is prompted to edit a plotting script to create figures for the paper using Python. The AI Scientist makes a note describing what each plot contains, enabling the saved figures and experimental notes to provide all the information required to write up the paper. At all steps, Aider sees its history of execution.

Note that, in general, the provided initial seed plotting and experiment templates are small, self-contained files. The AI Scientist frequently implements entirely new plots and collects new metrics that are not in the seed templates. This ability to arbitrarily edit the code occasionally leads to unexpected outcomes (Section 8).

3. Paper Write-up. The third phase of The AI Scientist produces a concise and informative write-up of its progress in the style of a standard machine learning conference proceeding in LaTeX. We note that writing good LaTeX can even take competent human researchers some time, so we take several steps to robustify the process. This consists of the following:

(a) **Per-Section Text Generation:** The recorded notes and plots are passed to Aider, which is prompted to fill in a blank conference template section by section. This goes in order of introduction, background, methods, experimental setup, results, and then the conclusion (all sections apart from the related work). All previous sections of the paper it has already written are in the context of the language model. We include brief tips and guidelines on what each section should include, based on the popular "How to ML Paper" guide, and include details in Appendix A.3. At each step of writing, Aider is prompted to *only use real experimental results in the form of notes and figures generated from code, and real citations* to reduce hallucination. Each section is initially refined with one round of self-reflection (Shinn et al., 2024) as it is being written. Aider is prompted to not include any citations in the text at this stage, and fill in only a skeleton for the related work, which will be completed in the next stage.

(b) **Web Search for References:** In a similar vein to idea generation, The AI Scientist is allowed 20 rounds to poll the Semantic Scholar API looking for the most relevant sources to compare and contrast the near-completed paper against for the related work section. This process also allows The AI Scientist to select any papers it would like to discuss and additionally fill in any citations that are missing from other sections of the paper. Alongside each selected paper, a short description is produced of where and how to include the citation, which is then passed to Aider. The paper's bibtex is automatically appended to the LaTeX file to guarantee correctness.

(c) **Refinement:** After the previous two stages, The AI Scientist has a completed first draft, but can often be overly verbose and repetitive. To resolve this, we perform one final round of self-reflection section-by-section, aiming to remove any duplicated information and streamline the arguments of the paper.

(d) **Compilation:** Once the LaTeX template has been filled in with all the appropriate results, this is fed into a LaTeX compiler. We use a LaTeX linter and pipe compilation errors back into Aider so that it can automatically correct any issues.

4. Automated Paper Reviewing

An LLM Reviewer Agent. A key component of an effective scientific community is its reviewing system, which evaluates and improves the quality of scientific papers. To mimic such a process using large language models, we design a GPT-4o-based agent (OpenAI, 2023) to conduct paper reviews based on the Neural Information Processing Systems (NeurIPS) conference review guidelines. The review agent processes the raw text of the PDF manuscript using the PyMuPDF parsing library. The output contains numerical scores (soundness, presentation, contribution, overall, confidence), lists of weaknesses and strengths as well as a preliminary binary decision (*accept* or *reject*). These decisions

may then be post-calibrated by thresholding using the reviewer score. We leverage this automated reviewing process to obtain an initial evaluation of the papers generated by THE AI SCIENTIST. We provide the entire reviewing prompt template in Appendix A.4.

Table 1 | **Performance of THE AI SCIENTIST's automated LLM reviewing system on 500 ICLR 2022 papers. We show mean and 95% bootstrap confidence intervals, and highlight the comparison between the human baseline and our best AI reviewer.**

	Reviewer	Balanced Acc. ↑	Accuracy ↑	F1 Score ↑	AUC ↑	FPR ↓	FNR ↓
	Human (NeurIPS)[1]	**0.66**	**0.73**	**0.49**	**0.65**	**0.17**	**0.52**
	Random Decision	0.50	0.50	0.40	0.50	0.50	0.50
	Always Reject	0.50	0.59	0.00	0.50	0.00	1.00
Uncalibrated	`Sonnet 3.5`	0.52 ± 0.01	0.40 ± 0.01	0.55 ± 0.01	0.52 ± 0.01	0.95 ± 0.02	0.00 ± 0.00
	`GPT-4o-mini`	0.53 ± 0.02	0.65 ± 0.01	0.11 ± 0.06	0.53 ± 0.02	0.01 ± 0.01	0.94 ± 0.04
	`GPT-4o` (0-shot)	0.61 ± 0.04	0.68 ± 0.03	0.43 ± 0.07	0.61 ± 0.04	0.11 ± 0.03	0.67 ± 0.07
	`GPT-4o` (1-shot)	0.60 ± 0.03	**0.70 ± 0.03**	0.37 ± 0.08	0.60 ± 0.03	0.04 ± 0.02	0.76 ± 0.06
Calibrated	`Sonnet 3.5` @8	0.59 ± 0.04	0.65 ± 0.04	0.45 ± 0.06	0.59 ± 0.04	0.20 ± 0.04	0.61 ± 0.07
	`GPT-4o-mini` @6	0.59 ± 0.04	0.64 ± 0.04	0.45 ± 0.06	0.59 ± 0.04	0.22 ± 0.05	0.60 ± 0.07
	`GPT-4o` (0-shot) @6	0.63 ± 0.04	0.63 ± 0.04	0.56 ± 0.05	0.63 ± 0.04	0.38 ± 0.05	0.36 ± 0.07
	`GPT-4o` (1-shot) @6	**0.65 ± 0.04**	**0.66 ± 0.04**	**0.57 ± 0.05**	**0.65 ± 0.04**	**0.31 ± 0.05**	**0.39 ± 0.07**

Evaluating the Automated Reviewer. To evaluate the LLM-based reviewer's performance, we compared the artificially generated decisions with ground truth data for 500 ICLR 2022 papers extracted from the publicly available OpenReview dataset (Berto, 2024). Similar to the previous section, we combine many recent advancements in LLM agents to make the decision-making process robust. More specifically, we improve the base LLM's decision-making process by leveraging self-reflection (Shinn et al., 2024), providing few-shot examples (Wei et al., 2022) and response ensembling (Wang et al., 2022). With GPT-4o, THE AI SCIENTIST's reviewing procedure achieves 70% accuracy when combining 5 rounds of self-reflection, 5 ensembled reviews, and a 1-shot review example taken from the ICLR 2022 review guidelines. Afterward, we perform an LLM-based meta-review, which prompts the agent to act as an Area Chair (Wang et al., 2022) (full prompts in Appendix A.4). While this number is lower than the 73% accuracy that was reported for humans in the NeurIPS 2021 consistency experiment (Beygelzimer et al., 2021), the automated reviewer achieves superhuman F1 Scores (0.57 vs. 0.49) and human-level AUC (0.65 for both) when thresholding the decision at a score of 6 (a "Weak Accept" in the NeurIPS review guidelines). This choice corresponds roughly to the average score of accepted papers.

The considered ICLR 2022 paper dataset is very class-imbalanced, i.e. it contains many more rejected papers. When considering a balanced dataset of papers, THE AI SCIENTIST's reviewing process achieves human-level accuracy (0.65% vs. 0.66%). Furthermore, the False Negative Rate (FNR) is much lower than the human baseline (0.39 vs. 0.52). Hence, the LLM-based review agent rejects fewer high-quality papers. The False Positive Rate (FNR), on the other hand, is higher (0.31 vs. 0.17) highlighting room for potential future improvements.

To further validate the performance of the automated reviewer, we compare the consistency of the overall paper scores between anonymous OpenReview reviewers randomly sampled pairwise per paper (Figure 2, bottom-left) and between the average of all reviewers and the LLM score (Figure 2, bottom-middle). For the set of 500 ICLR 2022 papers, we find that the correlation between the score of two human reviewers is smaller (0.14) than the correlation between the LLM score and the average score across the reviewers (0.18). Overall, across all metrics, the results suggest that LLM-based reviews can not only provide valuable feedback (D'Arcy et al., 2024) but also align more closely with the average human reviewer score than individual human reviewers align with each other.

[1]Numbers are calculated based of the NeurIPS consistency experiment (Beygelzimer et al., 2021).